**RAJARATA UNIVERSITY OF SRI LANKA**
**FACULTY OF APPLIED SCIENCES**

**B.Sc. in Information Technology**
**First Year - Semester I Examination – July / August 2023**

# ICT 1402 – PRINCIPLES OF PROGRAM DESIGN AND PROGRAMMING (PRACTICAL)

**Time: Two (02) hours**

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- This paper has four (04) questions in two (02) pages.
- Answer **ALL** questions.
- The total mark allotted for this paper is 50.
- Note that questions appear on both sides of the paper.
- If a page is not printed, please inform the supervisor immediately.

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1. Write a function to swap the values in two variables using the following function prototype.

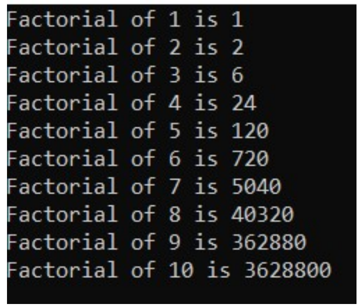
```
void swap(int *x, int *y);
```

Save the source file as Q1.c (10 Marks)

2. Write a **Recursive Function** to find the factorial of integers using the following function prototype.

```
int factorial(unsigned int);
```

Then, show how to use it to find the factorials of integers starting from 1 to 20.
The expected output of the program should appear as shown in Figure 1.
Save the source file as Q2.c (07 Marks)



```
Factorial of 1 is 1
Factorial of 2 is 2
Factorial of 3 is 6
Factorial of 4 is 24
Factorial of 5 is 120
Factorial of 6 is 720
Factorial of 7 is 5040
Factorial of 8 is 40320
Factorial of 9 is 362880
Factorial of 10 is 3628800
```

Figure 1: Expected output of the factorial program

3. Matrix shown in Figure 2 depicts a set of values stored in a patch of a digital image (size of 6×6) (**image1**). Those stored values are ranging from 0 to 255. It is required to **update** and **display** another image of the same dimension (**image2**) after reading the values in image1 based on the following condition.

   Condition: If the cell value in Figure 1 is greater than 50 and less than or equal to 110, update the corresponding cell in Figure 2 as 1, otherwise 0.

   Assume that the matrix used to represent image 2 has been initialized to zero when it is declared. Use **inner loops** to write the program and specify the print() statement within the loops.
   Save the source file as Q3.c (16 Marks)

| | | | | | |
|---|---|---|---|---|---|
| 50 | 50 | 20 | 30 | 240 | 130 |
| 50 | 20 | 100 | 105 | 140 | 120 |
| 40 | 105 | 20 | 20 | 105 | 120 |
| 20 | 100 | 20 | 10 | 110 | 135 |
| 10 | 200 | 100 | 105 | 110 | 20 |
| 0 | 250` | 200 | 120 | 111 | 0 |

Figure 2 – Matrix 1 represents the patch of the digital image 1.

4. Develop a C program to print tomorrow's date when the user enters the current date. Follow the instructions given below when implementing the program.
   Save your source code as Q4.c.
   Enter the current date based on the following format: mm dd yyyy
   Create a C structure named date to hold three variables, month, date, and year.
   Use the following array representing the days per month (starting from January to December) for the calculation.

```
const int daysPerMonth[12] = { 31, 28, 31, 30, 31,
                               30,31, 31, 30, 31, 30, 31 };
```

(17 marks)

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